

MATH 3260 - Chapter 7 Notes: Eigenvalues and Eigenvectors

Feedback-Revised Student Notes

Kennesaw State University - MATH 3260

Fall 2026 revision

Contents

Chapter overview	1
Section 7.1 - Eigenvalues and Eigenvectors	2
Learning objectives	2
Why eigenvectors are special	2
Detailed 2×2 eigenvalue calculation	4
Algebraic and geometric multiplicity	5
Eigenvalues of a linear transformation on P_2	6
Definitions and terminology	7
Section 7.2 - Diagonalization	8
Learning objectives	8
Why diagonalization is useful	8
Complete 3×3 diagonalization	10
Computing powers by diagonalization	11
Diagonalization of a linear transformation	12
Definitions and terminology	13
Section 7.3 - Symmetric Matrices and Orthogonal Diagonalization	14
Learning objectives	14
Detailed orthogonal diagonalization	16
Repeated eigenvalue and Gram-Schmidt	18
Definitions and terminology	18
Section 7.4 - Applications of Eigenvalues and Eigenvectors	20
Learning objectives	20
Population growth and stable age distributions	20
Systems of linear differential equations	21
Detailed rotation of a conic	23
Classification by eigenvalue signs	24
Definitions and terminology	26
Chapter 7 summary	26
Practical summary of algebraic and geometric multiplicity	26

These notes are synchronized with the feedback-revised student notebook. They emphasize definitions, theorems, worked mathematics, conceptual connections, and interpretation while omitting notebook-specific code and worksheet instructions.

Chapter overview

1. **Section 7.1:** Eigenvalues, eigenvectors, eigenspaces, and characteristic equations

2. **Section 7.2:** Similarity and diagonalization
3. **Section 7.3:** Symmetric matrices and orthogonal diagonalization
4. **Section 7.4:** Population models, systems of differential equations, quadratic forms, principal axes, and constrained optimization

Each section concludes with a terminology table, a worksheet, and complete instructor solutions.

Original-example policy

All numerical examples are original. Themes are used selectively from the San Antonio Spurs, Backstreet Boys, hiking, badminton, driving, travel, and R&B music. Fictional populations, listening profiles, performance data, and routes are labeled as teaching models.

Section 7.1 - Eigenvalues and Eigenvectors

Learning objectives

Students should be able to:

- explain the geometric meaning of an eigenvector;
- verify an eigenvalue-eigenvector pair;
- prove that an eigenspace is a subspace;
- derive and use the characteristic equation;
- compute eigenvalues and bases for eigenspaces;
- distinguish algebraic multiplicity from geometric multiplicity;
- find eigenvalues of triangular matrices;
- interpret eigenvalues and eigenvectors of a linear transformation.

Definition: eigenvalue and eigenvector

Let A be an $n \times n$ matrix. A scalar λ is an eigenvalue of A when there is a nonzero vector $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$. The nonzero vector $\mathbf{v}$ is an eigenvector corresponding to λ .

Why eigenvectors are special

A general vector may change both direction and length under multiplication by A . An eigenvector changes only by a scalar factor:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

- If $\lambda > 1$, the vector is stretched.
- If $0 < \lambda < 1$, it is contracted.
- If $\lambda < 0$, its direction reverses and its length is scaled by $|\lambda|$.
- If $\lambda = 0$, the vector is sent to the zero vector.

The line through an eigenvector is therefore an invariant direction of the transformation.

Why $\mathbf{v} \neq \mathbf{0}$ and where $A - \lambda I$ comes from

An eigenvector satisfies

$$A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}.$$

The zero vector is excluded because it satisfies the equation for every λ and identifies no special direction.

Rearrange:

$$A\mathbf{v} - \lambda\mathbf{v} = (A - \lambda I)\mathbf{v} = \mathbf{0}.$$

Thus the eigenvectors for a fixed λ are the nonzero vectors in $\ker(A - \lambda I)$. The eigenspace includes those vectors together with $\mathbf{0}$ so that it forms a subspace.

Spurs-inspired verification example

Consider the fictional matrix $A = \begin{bmatrix} 20 & 1 \\ 1 & 20 \end{bmatrix}$. The numbers are selected from familiar Spurs jersey values; this is not a real dataset.

Let

$$A = \begin{bmatrix} 20 & 1 \\ 1 & 20 \end{bmatrix}, \quad \mathbf{v}_+ = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_- = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Then

$$A\mathbf{v}_+ = \begin{bmatrix} 21 \\ 21 \end{bmatrix} = 21\mathbf{v}_+,$$

so $\mathbf{v}_+$ is an eigenvector with eigenvalue 21. Similarly,

$$A\mathbf{v}_- = \begin{bmatrix} 19 \\ -19 \end{bmatrix} = 19\mathbf{v}_-.$$

The two eigendirections are the lines $y = x$ and $y = -x$.

Definition: eigenspace

For an eigenvalue λ , the eigenspace is $E_\lambda = \ker(A - \lambda I)$. It contains all eigenvectors corresponding to λ together with the zero vector.

Theorem: an eigenspace is a subspace

If λ is an eigenvalue of A , then $E_\lambda = \ker(A - \lambda I)$ is a subspace of $\mathbb{R}^n$.

Complete proof

We verify the subspace conditions.

1. The zero vector belongs to the set.

$$(A - \lambda I)\mathbf{0} = \mathbf{0},$$

so $\mathbf{0} \in E_\lambda$.

2. Closure under addition. Suppose $\mathbf{u}, \mathbf{v} \in E_\lambda$. Then

$$(A - \lambda I)\mathbf{u} = \mathbf{0}, \quad (A - \lambda I)\mathbf{v} = \mathbf{0}.$$

Using distributivity,

$$\begin{aligned}
(A - \lambda I)(\mathbf{u} + \mathbf{v}) &= (A - \lambda I)\mathbf{u} + (A - \lambda I)\mathbf{v} \\
&= \mathbf{0} + \mathbf{0} \\
&= \mathbf{0}.
\end{aligned}$$

Therefore, $\mathbf{u} + \mathbf{v} \in E_\lambda$.

3. Closure under scalar multiplication. Suppose $\mathbf{u} \in E_\lambda$ and c is a scalar. Then

$$(A - \lambda I)(c\mathbf{u}) = c(A - \lambda I)\mathbf{u} = c\mathbf{0} = \mathbf{0}.$$

Thus $c\mathbf{u} \in E_\lambda$. All subspace conditions hold, so E_λ is a subspace. $\square$

Characteristic-equation criterion

A scalar λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$. Equivalently, one may use $\det(\lambda I - A) = 0$; the roots are the same.

Derivation and proof

Starting from the eigenvalue equation,

$$A\mathbf{v} = \lambda\mathbf{v},$$

move all terms to one side:

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

We require a **nonzero** solution $\mathbf{v}$. A homogeneous system $M\mathbf{v} = \mathbf{0}$ has a nonzero solution exactly when M is singular. A square matrix is singular exactly when its determinant is zero. Therefore,

$$(A - \lambda I)\mathbf{v} = \mathbf{0} \text{ has a nonzero solution} \iff \det(A - \lambda I) = 0.$$

This determinant is a polynomial in λ , called the characteristic polynomial. Its roots are the eigenvalues.

Detailed 2×2 eigenvalue calculation

Let

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic polynomial is

$$\begin{aligned}
p(\lambda) &= \det(A - \lambda I) \\
&= \det \begin{bmatrix} 4 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix} \\
&= (4 - \lambda)(3 - \lambda) - 2 \\
&= \lambda^2 - 7\lambda + 10 \\
&= (\lambda - 5)(\lambda - 2).
\end{aligned}$$

Hence the eigenvalues are $\lambda = 5$ and $\lambda = 2$.

Eigenspace for $\lambda = 5$

Solve $(A - 5I)\mathbf{v} = \mathbf{0}$:

$$A - 5I = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

The equation is

$$x - 2y = 0,$$

so $x = 2y$. Let $y = t$. Then

$$\mathbf{v} = t \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

and

$$E_5 = \text{span} \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}.$$

Eigenspace for $\lambda = 2$

Solve $(A - 2I)\mathbf{v} = \mathbf{0}$:

$$A - 2I = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Thus $x + y = 0$, so $x = -y$. Let $y = t$. Then

$$\mathbf{v} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

and

$$E_2 = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}.$$

Algebraic and geometric multiplicity

The **algebraic multiplicity** of an eigenvalue is its multiplicity as a root of the characteristic polynomial. The **geometric multiplicity** is the dimension of its eigenspace.

Consider

$$B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Because B is triangular, the characteristic polynomial is

$$(2 - \lambda)^3.$$

Thus $\lambda = 2$ has algebraic multiplicity 3. However,

$$B - 2I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

so $x_3 = 0$ while x_1 and x_2 are free. Therefore,

$$E_2 = \text{span}\{(1, 0, 0), (0, 1, 0)\},$$

and the geometric multiplicity is 2.

The **algebraic multiplicity** counts how many times λ occurs as a root of the characteristic polynomial. The **geometric multiplicity** is

$$\dim \ker(A - \lambda I).$$

Always,

$$1 \leq \text{geometric multiplicity} \leq \text{algebraic multiplicity}.$$

Theorem: eigenvalues of a triangular matrix

The eigenvalues of a triangular matrix are its diagonal entries, counted with algebraic multiplicity.

Proof

For an upper triangular matrix A , the matrix $A - \lambda I$ is also upper triangular. Its diagonal entries are

$$a_{11} - \lambda, a_{22} - \lambda, \dots, a_{nn} - \lambda.$$

The determinant of a triangular matrix is the product of its diagonal entries. Hence

$$\det(A - \lambda I) = \prod_{j=1}^n (a_{jj} - \lambda).$$

This product equals zero exactly when $\lambda = a_{jj}$ for at least one j . Therefore the eigenvalues are the diagonal entries. The same argument applies to lower triangular matrices. $\square$

Eigenvalues of a linear transformation on P_2

Define

$$T : P_2 \rightarrow P_2, \quad T(p) = p + xp'.$$

For $p(x) = a + bx + cx^2$,

$$p'(x) = b + 2cx,$$

so

$$T(p) = a + 2bx + 3cx^2.$$

Relative to the standard basis $\{1, x, x^2\}$, the matrix is

$$[T] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Therefore, the eigenvalues are 1, 2, 3, with eigenspaces

$$E_1 = \text{span}\{1\}, \quad E_2 = \text{span}\{x\}, \quad E_3 = \text{span}\{x^2\}.$$

Useful consequence

0 is an eigenvalue of A if and only if A is singular. Indeed, $A\mathbf{v} = 0\mathbf{v}$ for some nonzero $\mathbf{v}$ exactly when $A\mathbf{v} = \mathbf{0}$ has a nontrivial solution.

Definitions and terminology

Term	Meaning
Eigenvalue	Scalar λ for which $A\mathbf{v} = \lambda\mathbf{v}$ has a nonzero solution.
Eigenvector	Nonzero vector satisfying the eigenvalue equation.
Eigenspace	$E_\lambda = \ker(A - \lambda I)$.
Characteristic polynomial	$p(\lambda) = \det(A - \lambda I)$.
Characteristic equation	$\det(A - \lambda I) = 0$.
Algebraic multiplicity	Multiplicity of an eigenvalue as a polynomial root.
Geometric multiplicity	Dimension of the eigenspace.
Invariant direction	A line mapped into itself by the transformation.

Section 7.2 - Diagonalization

Learning objectives

Students should be able to:

- explain similarity and diagonalization;
- prove that similar matrices have the same characteristic polynomial;
- prove the diagonalization criterion;
- prove that eigenvectors associated with distinct eigenvalues are independent;
- diagonalize a matrix when possible;
- determine why a matrix fails to be diagonalizable;
- use diagonalization to compute powers and simplify a linear transformation.

Definition: similarity and diagonalization

Matrices A and B are similar if $B = P^{-1}AP$ for some invertible P . A matrix A is diagonalizable if it is similar to a diagonal matrix D .

Why diagonalization is useful

If

$$A = PDP^{-1},$$

then

$$A^k = (PDP^{-1})^k = PD^kP^{-1}.$$

All middle factors $P^{-1}P$ cancel. Since powers of a diagonal matrix are computed entry by entry, repeated applications of A become much easier.

Theorem: similar matrices have the same characteristic polynomial

If $B = P^{-1}AP$, then $\det(\lambda I - B) = \det(\lambda I - A)$. Therefore A and B have the same eigenvalues with the same algebraic multiplicities.

Complete proof

Assume $B = P^{-1}AP$. Because $P^{-1}IP = I$,

$$\begin{aligned}\lambda I - B &= \lambda P^{-1}IP - P^{-1}AP \\ &= P^{-1}(\lambda I - A)P.\end{aligned}$$

Taking determinants and using $\det(XY) = \det(X)\det(Y)$,

$$\begin{aligned}\det(\lambda I - B) &= \det(P^{-1})\det(\lambda I - A)\det(P) \\ &= \frac{1}{\det(P)}\det(\lambda I - A)\det(P) \\ &= \det(\lambda I - A).\end{aligned}$$

Thus the characteristic polynomials are identical, so their roots and multiplicities are identical. $\square$

Theorem: diagonalization criterion

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

Proof: independent eigenvectors imply diagonalization

Suppose A has linearly independent eigenvectors $\mathbf{p}_1, \dots, \mathbf{p}_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. Form

$$P = [\mathbf{p}_1 \ \mathbf{p}_2 \ \cdots \ \mathbf{p}_n], \quad D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Because the columns of P are linearly independent, P is invertible. Moreover,

$$AP = [A\mathbf{p}_1 \ \cdots \ A\mathbf{p}_n] = [\lambda_1\mathbf{p}_1 \ \cdots \ \lambda_n\mathbf{p}_n] = PD.$$

Multiplying on the left by P^{-1} gives

$$P^{-1}AP = D.$$

Therefore, A is diagonalizable.

Proof: diagonalization implies independent eigenvectors

Suppose $P^{-1}AP = D$, where D is diagonal and P is invertible. Rearranging gives

$$AP = PD.$$

Write the columns of P as $\mathbf{p}_1, \dots, \mathbf{p}_n$, and let the diagonal entries of D be $\lambda_1, \dots, \lambda_n$. Comparing columns in $AP = PD$ yields

$$A\mathbf{p}_j = \lambda_j\mathbf{p}_j$$

for every j . Thus every column of P is an eigenvector. Because P is invertible, its columns are linearly independent. Hence A has n linearly independent eigenvectors. $\square$

Theorem: distinct eigenvalues give independent eigenvectors

Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Proof by induction

The statement is immediate for one eigenvector. Assume it is true for $k - 1$ eigenvectors. Let

$$A\mathbf{v}_j = \lambda_j\mathbf{v}_j, \quad \lambda_1, \dots, \lambda_k \text{ distinct,}$$

and suppose

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}.$$

Apply A :

$$c_1\lambda_1\mathbf{v}_1 + \cdots + c_k\lambda_k\mathbf{v}_k = \mathbf{0}.$$

Multiply the original relation by λ_k and subtract:

$$c_1(\lambda_1 - \lambda_k)\mathbf{v}_1 + \cdots + c_{k-1}(\lambda_{k-1} - \lambda_k)\mathbf{v}_{k-1} = \mathbf{0}.$$

By the induction hypothesis, $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ are independent. Since every $\lambda_j - \lambda_k \neq 0$, we obtain $c_1 = \cdots = c_{k-1} = 0$. The original relation then gives $c_k\mathbf{v}_k = \mathbf{0}$, and because $\mathbf{v}_k \neq \mathbf{0}$, $c_k = 0$. Thus all coefficients are zero. $\square$

Complete 3×3 diagonalization

Consider

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix}.$$

The characteristic polynomial factors as

$$\det(\lambda I - A) = (\lambda - 1)(\lambda - 2)(\lambda - 3).$$

The eigenvalues are distinct, so the corresponding eigenvectors will automatically be linearly independent. We still compute each eigenspace explicitly.

Eigenvalue $\lambda = 1$

$$A - I = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$x_1 + x_3 = 0, \quad x_2 = 0.$$

Let $x_3 = t$. Then

$$\mathbf{x} = t(-1, 0, 1),$$

so choose

$$\mathbf{p}_1 = (1, 0, -1)$$

as an equivalent eigenvector.

Eigenvalue $\lambda = 2$

$$A - 2I = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence

$$x_1 = -x_3, \quad x_2 = x_3.$$

Let $x_3 = t$. Then

$$\mathbf{x} = t(-1, 1, 1),$$

so choose

$$\mathbf{p}_2 = (-1, 1, 1).$$

Eigenvalue $\lambda = 3$

$$A - 3I = \begin{bmatrix} -1 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & 1 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$x_1 = -x_2, \quad x_3 = 0.$$

Let $x_2 = t$. Then

$$\mathbf{x} = t(-1, 1, 0),$$

so choose

$$\mathbf{p}_3 = (-1, 1, 0).$$

Form

$$P = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Because $\det(P) = -1$, the matrix P is invertible. Its inverse is

$$P^{-1} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix}.$$

Direct multiplication gives

$$P^{-1}AP = D.$$

Computing powers by diagonalization

Because $A = PDP^{-1}$,

$$A^5 = PD^5P^{-1},$$

where

$$D^5 = \text{diag}(1, 32, 243).$$

The result is

$$A^5 = \begin{bmatrix} 212 & -31 & 211 \\ -211 & 32 & -211 \\ 31 & 31 & 32 \end{bmatrix}.$$

This avoids four full matrix multiplications.

If

$$A = PDP^{-1},$$

then

$$A^5 = PD^5P^{-1},$$

and more generally $A^k = PD^kP^{-1}$.

A matrix that is not diagonalizable

The matrix $J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ has only one independent eigenvector, so it is not diagonalizable.

Indeed, the only eigenvalue is $\lambda = 2$, with algebraic multiplicity 2. But

$$J - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

so $y = 0$ and

$$E_2 = \text{span}\{(1, 0)\}.$$

The geometric multiplicity is 1, smaller than the algebraic multiplicity 2. Therefore, there are not two linearly independent eigenvectors.

Diagonalization of a linear transformation

If a linear transformation $T : V \rightarrow V$ has a basis

$$B = \{\mathbf{p}_1, \dots, \mathbf{p}_n\}$$

consisting of eigenvectors, then the matrix of T relative to B is diagonal:

$$[T]_B = \text{diag}(\lambda_1, \dots, \lambda_n).$$

The diagonal entries record the independent scaling of each eigenvector direction.

Definitions and terminology

Term	Meaning
Similar matrices	$B = P^{-1}AP$ for an invertible P .
Diagonalizable	Similar to a diagonal matrix.
Eigenbasis	Basis consisting entirely of eigenvectors.
Diagonalization	Factorization $A = PDP^{-1}$.
Defective matrix	Matrix lacking enough independent eigenvectors.
Algebraic multiplicity	Multiplicity in the characteristic polynomial.
Geometric multiplicity	Dimension of the eigenspace.

Section 7.3 - Symmetric Matrices and Orthogonal Diagonalization

Learning objectives

Students should be able to:

- recognize symmetric and orthogonal matrices;
- prove the column criterion for orthogonal matrices;
- prove that eigenvectors of a symmetric matrix associated with distinct eigenvalues are orthogonal;
- understand why the eigenvalues of a real symmetric matrix are real;
- apply the spectral theorem;
- orthogonally diagonalize symmetric matrices, including repeated-eigenvalue cases.

Definition: symmetric matrix

A real square matrix is symmetric when $A^T = A$. Thus $a_{ij} = a_{ji}$ for all entries.

Definition: orthogonal matrix

A real square matrix P is orthogonal when $P^T P = I$. Equivalently, $P^{-1} = P^T$.

Theorem: column criterion for orthogonality

An $n \times n$ matrix P is orthogonal if and only if its columns form an orthonormal set.

Complete proof

Write

$$P = [\mathbf{p}_1 \ \cdots \ \mathbf{p}_n].$$

The (i, j) entry of $P^T P$ is

$$(P^T P)_{ij} = \mathbf{p}_i^T \mathbf{p}_j = \mathbf{p}_i \cdot \mathbf{p}_j.$$

Therefore,

$$P^T P = I$$

exactly when

$$\mathbf{p}_i \cdot \mathbf{p}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

These are precisely the conditions that the columns form an orthonormal set. $\square$

Theorem: distinct eigenvalues of a symmetric matrix have orthogonal eigenvectors

If $A = A^T$, $A\mathbf{x} = \lambda\mathbf{x}$, $A\mathbf{y} = \mu\mathbf{y}$, and $\lambda \neq \mu$, then $\mathbf{x} \cdot \mathbf{y} = 0$.

Complete proof

Because A is symmetric,

$$\mathbf{x}^T A \mathbf{y} = (A \mathbf{x})^T \mathbf{y}.$$

Using the eigenvalue equations,

$$\mathbf{x}^T A \mathbf{y} = \mathbf{x}^T (\mu \mathbf{y}) = \mu \mathbf{x}^T \mathbf{y},$$

while

$$(A \mathbf{x})^T \mathbf{y} = (\lambda \mathbf{x})^T \mathbf{y} = \lambda \mathbf{x}^T \mathbf{y}.$$

Thus

$$\mu \mathbf{x}^T \mathbf{y} = \lambda \mathbf{x}^T \mathbf{y},$$

so

$$(\lambda - \mu) \mathbf{x}^T \mathbf{y} = 0.$$

Since $\lambda \neq \mu$, it follows that

$$\mathbf{x}^T \mathbf{y} = 0.$$

Therefore the eigenvectors are orthogonal. $\square$

Why symmetric matrices have real eigenvalues

Every eigenvalue of a real symmetric matrix is real.

Detailed proof using complex vectors

A real matrix can have complex eigenvalues, so temporarily allow a complex eigenvector $\mathbf{z} \neq 0$ with

$$A \mathbf{z} = \lambda \mathbf{z}.$$

Let $\mathbf{z}^*$ denote the conjugate transpose. Multiply by $\mathbf{z}^*$:

$$\mathbf{z}^* A \mathbf{z} = \lambda \mathbf{z}^* \mathbf{z}.$$

The scalar $\mathbf{z}^* \mathbf{z}$ is positive and real. We now show that $\mathbf{z}^* A \mathbf{z}$ is real. Taking its complex conjugate gives

$$\overline{\mathbf{z}^* A \mathbf{z}} = \mathbf{z}^* A^T \mathbf{z}.$$

Because $A^T = A$,

$$\overline{\mathbf{z}^* A \mathbf{z}} = \mathbf{z}^* A \mathbf{z}.$$

A scalar equal to its own complex conjugate is real. Therefore,

$$\lambda = \frac{\mathbf{z}^* A \mathbf{z}}{\mathbf{z}^* \mathbf{z}}$$

is real. $\square$

Spectral theorem

A real matrix is orthogonally diagonalizable if and only if it is symmetric. Thus, for symmetric A , there exists an orthogonal P and a real diagonal D such that $A = PDP^T$.

Proof framework

The full result can be organized in two directions.

If A is orthogonally diagonalizable, then A is symmetric. Suppose

$$A = PDP^T,$$

where P is orthogonal and D is diagonal. Then

$$A^T = (PDP^T)^T = PD^T P^T = PDP^T = A,$$

because $D^T = D$.

If A is symmetric, then A is orthogonally diagonalizable. The proof proceeds by induction on the dimension.

1. A symmetric matrix has a real eigenvalue and a corresponding unit eigenvector $\mathbf{q}_1$.
2. The orthogonal complement $\mathbf{q}_1^\perp$ is invariant under A . To see this, let $\mathbf{y} \perp \mathbf{q}_1$. Then

$$\mathbf{q}_1^T A \mathbf{y} = (A \mathbf{q}_1)^T \mathbf{y} = (\lambda_1 \mathbf{q}_1)^T \mathbf{y} = \lambda_1 \mathbf{q}_1^T \mathbf{y} = 0.$$

Hence $A \mathbf{y} \in \mathbf{q}_1^\perp$. 3. The restriction of A to $\mathbf{q}_1^\perp$ is again symmetric. By the induction hypothesis, it has an orthonormal eigenbasis. 4. Adjoin $\mathbf{q}_1$ to that basis. The result is an orthonormal eigenbasis for the entire space.

Using these eigenvectors as columns of P produces $P^T A P = D$. $\square$

Detailed orthogonal diagonalization

Let

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}.$$

The characteristic polynomial is

$$\begin{aligned} p(\lambda) &= \det(A - \lambda I) \\ &= (5 - \lambda)(2 - \lambda) - 4 \\ &= \lambda^2 - 7\lambda + 6 \\ &= (\lambda - 6)(\lambda - 1). \end{aligned}$$

Thus the eigenvalues are 6 and 1.

Along an eigendirection, a unit eigenvector is stretched or reflected only by the scalar λ , so its image has length $|\lambda|$.

Eigenvector for $\lambda = 6$

$$A - 6I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

Thus $x = 2y$. Choose

$$\mathbf{v}_1 = (2, 1).$$

Its norm is $\sqrt{5}$, so

$$\mathbf{q}_1 = \frac{1}{\sqrt{5}}(2, 1).$$

Eigenvector for $\lambda = 1$

$$A - I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}.$$

Thus $2x + y = 0$. Choose

$$\mathbf{v}_2 = (-1, 2).$$

Its norm is $\sqrt{5}$, so

$$\mathbf{q}_2 = \frac{1}{\sqrt{5}}(-1, 2).$$

Notice that

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 2(-1) + 1(2) = 0,$$

as predicted by the symmetric-matrix theorem.

Form

$$P = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix}.$$

The columns of P are orthonormal, so

$$P^{-1} = P^T.$$

Direct multiplication verifies

$$P^T A P = D.$$

Repeated eigenvalue and Gram-Schmidt

Consider

$$C = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

The eigenvalues are 4 and 1, where 1 has algebraic multiplicity 2.

For $\lambda = 4$, choose

$$\mathbf{v}_3 = (1, 1, 1), \quad \mathbf{q}_3 = \frac{1}{\sqrt{3}}(1, 1, 1).$$

For $\lambda = 1$, one possible basis is

$$\mathbf{v}_1 = (1, -1, 0), \quad \mathbf{v}_2 = (1, 0, -1).$$

These vectors are independent but not orthogonal because $\mathbf{v}_1 \cdot \mathbf{v}_2 = 1$.

Apply Gram-Schmidt within the eigenspace:

$$\mathbf{w}_1 = \mathbf{v}_1 = (1, -1, 0).$$

Next,

$$\begin{aligned} \mathbf{w}_2 &= \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1} \mathbf{w}_1 \\ &= (1, 0, -1) - \frac{1}{2}(1, -1, 0) \\ &= \left(\frac{1}{2}, \frac{1}{2}, -1\right). \end{aligned}$$

Scale by 2 to use the simpler vector

$$\tilde{\mathbf{w}}_2 = (1, 1, -2).$$

Normalize:

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \mathbf{q}_2 = \frac{1}{\sqrt{6}}(1, 1, -2).$$

Then $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$ is an orthonormal eigenbasis, and

$$P^T C P = \text{diag}(1, 1, 4).$$

Definitions and terminology

Term	Meaning
Symmetric matrix	$A^T = A$.
Orthogonal matrix	$P^T P = I$, so $P^{-1} = P^T$.
Orthogonal diagonalization	$P^T A P = D$ with P orthogonal and D diagonal.
Spectral theorem	Real symmetric matrices have orthonormal eigenbases.
Spectral decomposition	$A = P D P^T$ or $A = \sum \lambda_i \mathbf{q}_i \mathbf{q}_i^T$.
Repeated eigenspace	Eigenspace associated with an eigenvalue of multiplicity greater than one.

Section 7.4 - Applications of Eigenvalues and Eigenvectors

Learning objectives

Students should be able to:

- construct and analyze an age-transition population model;
- find and interpret a stable age distribution;
- solve a diagonalizable system $\mathbf{y}' = A\mathbf{y}$;
- represent a quadratic form by a symmetric matrix;
- use the principal-axes theorem to remove cross-product terms;
- classify quadratic forms using eigenvalues;
- prove and apply the constrained-extrema theorem.

Population growth and stable age distributions

Consider a fictional wildlife population near a hiking preserve. Divide the population into three age classes:

1. first-year,
2. second-year,
3. third-year.

Use the Leslie matrix

$$L = \begin{bmatrix} 0 & \frac{6}{5} & 2 \\ \frac{11}{20} & 0 & 0 \\ 0 & \frac{11}{20} & 0 \end{bmatrix}.$$

The first row contains fertility rates. The subdiagonal contains survival rates.

A stable age distribution is an eigenvector with nonnegative entries. Verify that

$$\mathbf{x} = \begin{bmatrix} 20 \\ 10 \\ 5 \end{bmatrix}$$

satisfies

$$L\mathbf{x} = \begin{bmatrix} 22 \\ 11 \\ 11/2 \end{bmatrix} = \frac{11}{10} \begin{bmatrix} 20 \\ 10 \\ 5 \end{bmatrix}.$$

Thus the growth factor is

$$\lambda = \frac{11}{10} = 1.1,$$

and the age ratio remains

$$20 : 10 : 5 = 4 : 2 : 1.$$

The corresponding percentages are

$$\frac{4}{7}, \quad \frac{2}{7}, \quad \frac{1}{7}.$$

Interpretation

The eigenvalue gives the long-run growth factor per transition, while the normalized positive eigenvector gives the long-run proportion in each age class.

Systems of linear differential equations

Consider the coupled system

$$\mathbf{y}' = A\mathbf{y}, \quad A = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix}.$$

This may be viewed as a fictional two-component badminton recovery model: each component decays but is weakly coupled to the other.

The eigenvalues are

$$\lambda_1 = -2, \quad \lambda_2 = -4,$$

with eigenvectors

$$\mathbf{p}_1 = (1, 1), \quad \mathbf{p}_2 = (1, -1).$$

Form

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}.$$

Set

$$\mathbf{y} = P\mathbf{w}.$$

Then

$$P\mathbf{w}' = AP\mathbf{w}.$$

Multiplying by P^{-1} gives

$$\mathbf{w}' = P^{-1}AP\mathbf{w} = D\mathbf{w}.$$

Therefore the coupled system separates into

$$w_1' = -2w_1, \quad w_2' = -4w_2.$$

Thus

$$w_1 = c_1 e^{-2t}, \quad w_2 = c_2 e^{-4t}.$$

Returning to the original variables,

$$\mathbf{y}(t) = c_1 e^{-2t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

If

$$\mathbf{y}(0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix},$$

then

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The equations are

$$c_1 + c_2 = 3, \quad c_1 - c_2 = 1.$$

Adding gives $2c_1 = 4$, so $c_1 = 2$ and $c_2 = 1$. Hence

$$\boxed{\mathbf{y}(t) = 2e^{-2t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{-4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}}.$$

Definition: quadratic form

A quadratic form in n variables is $q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$, where A may be taken symmetric.

For two variables,

$$q(x, y) = ax^2 + bxy + cy^2$$

has symmetric matrix

$$A = \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix},$$

because

$$\begin{aligned} \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= ax^2 + \frac{b}{2}xy + \frac{b}{2}yx + cy^2 \\ &= ax^2 + bxy + cy^2. \end{aligned}$$

Principal-axes theorem

If A is symmetric, choose an orthogonal matrix P such that $P^T A P = D$. Under the rotation $\mathbf{x} = P\mathbf{u}$, the quadratic form becomes $q = \mathbf{u}^T D \mathbf{u}$, so all cross terms disappear.

Proof

Let

$$q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

and use the orthogonal coordinate change

$$\mathbf{x} = P\mathbf{u}.$$

Then

$$\begin{aligned} q &= (P\mathbf{u})^T A (P\mathbf{u}) \\ &= \mathbf{u}^T P^T A P \mathbf{u} \\ &= \mathbf{u}^T D \mathbf{u}. \end{aligned}$$

Because D is diagonal,

$$q = \lambda_1 u_1^2 + \cdots + \lambda_n u_n^2.$$

No mixed products $u_i u_j$ remain. The new coordinate axes are precisely the orthonormal eigenvector directions. $\square$

Detailed rotation of a conic

Consider

$$5x^2 + 4xy + 2y^2 = 12.$$

The quadratic-form matrix is

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}.$$

From Section 7.3, an orthogonal diagonalization is

$$P = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}, \quad P^T A P = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix}.$$

Set

$$\begin{bmatrix} x \\ y \end{bmatrix} = P \begin{bmatrix} u \\ v \end{bmatrix}.$$

Then the equation becomes

$$6u^2 + v^2 = 12,$$

or

$$\frac{u^2}{2} + \frac{v^2}{12} = 1.$$

This is an ellipse whose principal axes are the eigenvector directions. The rotation angle satisfies

$$\cos \theta = \frac{2}{\sqrt{5}}, \quad \sin \theta = \frac{1}{\sqrt{5}},$$

so

$$\theta = \arctan \left(\frac{1}{2} \right).$$

Classification by eigenvalue signs

For $q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ with symmetric A :

- all eigenvalues positive: q is positive definite;
- all eigenvalues negative: q is negative definite;
- both positive and negative eigenvalues: q is indefinite;
- zero eigenvalues with no sign change: q is semidefinite.

For example,

$$4x^2 + y^2 + 9z^2 = 36$$

has three positive eigenvalues and describes an ellipsoid. In contrast,

$$x^2 + y^2 - z^2 = 1$$

has eigenvalues 1, 1, -1 and describes a one-sheet hyperboloid.

Constrained-extrema theorem

For a symmetric matrix A , the maximum of $q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ subject to $\|\mathbf{x}\| = 1$ is the largest eigenvalue; the minimum is the smallest eigenvalue.

Complete proof using the spectral theorem

Let

$$A = P D P^T,$$

where P is orthogonal and

$$D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Set

$$\mathbf{y} = P^T \mathbf{x}.$$

Because P is orthogonal,

$$\|\mathbf{y}\| = \|\mathbf{x}\| = 1.$$

Then

$$\begin{aligned}
q(\mathbf{x}) &= \mathbf{x}^T A \mathbf{x} \\
&= \mathbf{x}^T P D P^T \mathbf{x} \\
&= \mathbf{y}^T D \mathbf{y} \\
&= \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2.
\end{aligned}$$

Since

$$y_1^2 + \cdots + y_n^2 = 1,$$

the value of q is a weighted average of the eigenvalues, with nonnegative weights y_i^2 . Therefore,

$$\lambda_{\min} \leq q(\mathbf{x}) \leq \lambda_{\max}.$$

The maximum is achieved when $\mathbf{y}$ is a coordinate vector corresponding to $\lambda_{\max}$, which means $\mathbf{x}$ is a unit eigenvector for $\lambda_{\max}$. Similarly, the minimum occurs at a unit eigenvector for $\lambda_{\min}$. $\square$

For

$$q(x, y) = 5x^2 + 4xy + 2y^2, \quad x^2 + y^2 = 1,$$

the eigenvalues are 6 and 1. Therefore,

$$\max q = 6$$

at

$$\pm \frac{1}{\sqrt{5}}(2, 1),$$

and

$$\min q = 1$$

at

$$\pm \frac{1}{\sqrt{5}}(-1, 2).$$

Rayleigh quotient

For nonzero $\mathbf{x}$,

$$R_A(\mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}.$$

If $\mathbf{x}$ is an eigenvector with eigenvalue λ , then $R_A(\mathbf{x}) = \lambda$. For an approximate eigenvector it provides an eigenvalue estimate.

Definitions and terminology

Term	Meaning
Leslie matrix	Age-transition matrix containing fertility and survival rates.
Stable age distribution	Positive eigenvector giving constant long-run proportions.
Growth factor	Dominant positive eigenvalue of the population matrix.
Linear system $\mathbf{y}' = A\mathbf{y}$	Coupled first-order differential equations.
Quadratic form	$q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$.
Principal axes	Orthogonal eigenvector directions of a symmetric quadratic-form matrix.
Positive definite	All eigenvalues positive.
Rayleigh quotient	$\mathbf{x}^T A \mathbf{x} / (\mathbf{x}^T \mathbf{x})$.

Chapter 7 summary

The central idea of Chapter 7 is that a matrix may possess special directions that it only scales. These eigendirections reveal the long-term behavior of repeated transformations, permit diagonalization, explain the geometry of symmetric matrices, decouple differential equations, determine stable population structures, remove cross terms from quadratic forms, and solve constrained optimization problems.

A useful conceptual chain is

$$\text{eigenvectors} \implies \text{eigenbasis} \implies \text{diagonalization} \implies \text{simple powers and applications.}$$

For symmetric matrices, the spectral theorem strengthens this chain by guaranteeing an **orthonormal** eigenbasis.

Practical summary of algebraic and geometric multiplicity

For an eigenvalue λ ,

$$\text{algebraic multiplicity} = \text{multiplicity of } \lambda \text{ as a root of } \det(A - \lambda I),$$

while

$$\text{geometric multiplicity} = \dim \ker(A - \lambda I).$$

A matrix is diagonalizable exactly when it has enough linearly independent eigenvectors. Comparing the two multiplicities is therefore one of the fastest ways to diagnose diagonalizability.